

LQR with Optimal Feedback Control (OFC)

Mode: lqr-applied-to-cart-manip-following-cart

2D Cart-Pendulum System with Muscle Dynamics and Manipulator Tracking

Research Note

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1 Overview

This note describes the formulation and implementation of the `lqr-applied-to-cart-manip-following-cart` simulation mode. The system consists of two coupled sub-systems:

1. **Cart-Pendulum (4 DOF):** A 2D cart moving in the XY -plane carrying a 3D gimbal pendulum. It is controlled by a biologically inspired Optimal Feedback Control (OFC) loop using Muscle Dynamics, a Zero-Force-Trajectory (ZFT) Reference Mass, Impedance Force, and a Finite-Horizon LQR.
2. **Cup Manipulator (2 DOF):** A 2-link revolute manipulator whose end-effector (EE) is driven to track the cart's 2D position via online Inverse Kinematics (IK) and a Joint-Space Computed-Torque Controller.

The two sub-systems share the same Drake `MultibodyPlant` but have *separate model instances* so their states and actuations can be queried and applied independently.

2 State and Coordinate Definitions

2.1 Cart-Pendulum State (8D)

$$\mathbf{s}_{\text{cart}} = \underbrace{[x, y]^T}_{\text{cart position}} \oplus \underbrace{[\alpha, \beta]^T}_{\text{pend. angles}} \oplus \underbrace{[\dot{x}, \dot{y}]^T}_{\text{cart velocity}} \oplus \underbrace{[\dot{\alpha}, \dot{\beta}]^T}_{\text{pend. ang. vel.}} \in \mathbb{R}^8$$

where α is the pendulum pitch and β is the roll (gimbal angles).

2.2 Full OFC State (14D)

The LQR operates on the extended state:

$$\mathbf{x} = [x, y, \alpha, \beta, \dot{x}, \dot{y}, \dot{\alpha}, \dot{\beta}, F_x, F_y, x_{\text{ref}}, y_{\text{ref}}, \dot{x}_{\text{ref}}, \dot{y}_{\text{ref}}]^T \in \mathbb{R}^{14}$$

2.3 Manipulator State (4D)

$$\mathbf{s}_{\text{manip}} = [q_1, q_2, \dot{q}_1, \dot{q}_2]^T \in \mathbb{R}^4$$

where q_1 is the base-to-link1 joint angle and q_2 is the link1-to-link2 joint angle.

3 Control Architecture

3.1 Cart-Pendulum Control Loop (OFC)

The cart-pendulum control uses a biologically inspired OFC architecture with four cascaded blocks:

$$\text{LQR} \xrightarrow{u} \text{Muscle Dynamics} \xrightarrow{\mathbf{F}} \text{ZFT Reference Mass} \xrightarrow{\mathbf{s}_{\text{ref}}} \text{Impedance Force} \xrightarrow{\mathbf{F}_{\text{imp}}} \text{Cart-Pendulum Plant}$$

Important: the muscle force $\mathbf{F}$ is fed into the *ZFT reference mass*, not directly into the impedance block. The impedance block receives only $\mathbf{s}_{\text{ref}}$ (position and velocity of the reference mass) and $\mathbf{s}_{\text{cart}}$ (actual cart state). $\mathbf{F}$ reaches the impedance output only *indirectly*, by shaping how the reference mass moves.

Block 1 — Muscle Dynamics (Low-Pass Actuator Filter)

The neural command $\mathbf{u} = [u_x, u_y]^\top$ is filtered through first-order muscle dynamics:

$$\dot{F}_i = \frac{-F_i + u_i}{\tau_m}, \quad i \in \{x, y\}$$

- **Input:** $\mathbf{u} \in \mathbb{R}^2$ (LQR neural command)
- **State:** $\mathbf{F} = [F_x, F_y]^\top \in \mathbb{R}^2$
- **Output:** $\mathbf{F} \in \mathbb{R}^2$ (muscle force)
- **Time constant:** $\tau_m = 0.03$ s (human muscle)

This low-pass filter captures the finite bandwidth of biological muscles and prevents instantaneous force jumps.

Block 2 — ZFT Reference Mass (Smooth Reference Trajectory)

The muscle force $\mathbf{F}$ is the *only* direct input from the muscle block that propagates forward. It drives the ZFT reference mass ODE together with a spring-damper coupling to the actual cart position. This is the mechanism by which the neural command shapes the reference trajectory — $\mathbf{F}$ never enters the impedance block directly.

The ZFT dynamics are:

$$M_{\text{ref}} \ddot{\mathbf{p}}_{\text{ref}} = \mathbf{F} + K(\mathbf{p} - \mathbf{p}_{\text{ref}}) + D(\dot{\mathbf{p}} - \dot{\mathbf{p}}_{\text{ref}})$$

where $\mathbf{p} = [x, y]^\top$, $\mathbf{p}_{\text{ref}} = [x_{\text{ref}}, y_{\text{ref}}]^\top$.

- **Inputs:** $\mathbf{s}_{\text{cart}} = [x, y, \dot{x}, \dot{y}]^\top \in \mathbb{R}^4$ and $\mathbf{F} \in \mathbb{R}^2$
- **State:** $\mathbf{s}_{\text{ref}} = [x_{\text{ref}}, y_{\text{ref}}, \dot{x}_{\text{ref}}, \dot{y}_{\text{ref}}]^\top \in \mathbb{R}^4$
- **Output:** $\mathbf{s}_{\text{ref}} \in \mathbb{R}^4$
- **Parameters:** $M_{\text{ref}} = 1.0$ kg, $K = 50$ N/m, $D = 10$ N · s/m

The ZFT reference mass provides the “efference copy” of the intended trajectory, decoupling perception from control.

Block 3 — Impedance Force (Spring-Damper Coupling)

The impedance block computes the external force applied to the cart by connecting it to the ZFT reference via a virtual spring-damper:

$$\mathbf{F}_{\text{imp}} = K(\mathbf{p}_{\text{ref}} - \mathbf{p}) + D(\dot{\mathbf{p}}_{\text{ref}} - \dot{\mathbf{p}})$$

- **Inputs:** $\mathbf{s}_{\text{cart}} \in \mathbb{R}^4$ (actual cart pos. + vel.) and $\mathbf{s}_{\text{ref}} \in \mathbb{R}^4$ (ZFT reference pos. + vel.).
The muscle force $\mathbf{F}$ is NOT a direct input here.
- **Output:** $\mathbf{F}_{\text{imp}} \in \mathbb{R}^2$ (cart actuation)
- **Parameters:** $K = 50$ N/m, $D = 10$ N · s/m

$\mathbf{F}_{\text{imp}}$ is applied directly to the cart’s prismatic actuators. The pendulum receives no direct torque input — it is shaped purely by the resulting cart motion. The muscle force $\mathbf{F}$ influences $\mathbf{F}_{\text{imp}}$ only *indirectly*: by driving the ZFT ODE, it changes $\mathbf{s}_{\text{ref}}$, which in turn changes the spring-damper error seen by the impedance block.

Block 4 — Finite-Horizon LQR Controller

The LQR minimises the quadratic cost over a finite horizon T :

$$J = \int_0^T \left[\mathbf{e}^\top Q \mathbf{e} + \mathbf{u}^\top R \mathbf{u} \right] dt + \mathbf{e}(T)^\top Q_N \mathbf{e}(T)$$

where $\mathbf{e} = \mathbf{x} - \mathbf{x}^*$ is the 14D state error with goal state

$$\mathbf{x}^* = [x^*, y^*, 0, 0, 0, 0, 0, 0, 0, 0, x^*, y^*, 0, 0]^\top$$

The time-varying gain is obtained by solving the backward Riccati equation:

$$-\dot{P}(t) = A^\top P + PA - PBR^{-1}B^\top P + Q, \quad P(T) = Q_N$$

yielding the control law:

$$\mathbf{u}(t) = -R^{-1}B^\top P(t) \mathbf{e}(t)$$

Cost matrices:

$$\begin{aligned} Q &= \text{diag}(100, 100, 500, 500, 10, 10, 100, 100, 0.1, 0.1, 1, 1, 0.1, 0.1) \\ Q_N &= 2Q \\ R &= \text{diag}(1, 1) \end{aligned}$$

The system (A, B) is obtained by linearising the 14D OFC dynamics around the equilibrium (upright pendulum, zero velocity, zero force, cart at goal).

3.2 Manipulator Control Loop

The manipulator tracks the cart's 2D position using two cascaded blocks:

$$\text{Cart Position} \xrightarrow{\mathbf{p}_{\text{cart}}} \text{IK Solver} \xrightarrow{[q_d, \dot{q}_d]} \text{Computed-Torque Controller} \xrightarrow{\boldsymbol{\tau}} \text{Manipulator Plant}$$

Block 5 — Inverse Kinematics (IK Solver)

At every control step the desired cart position $\mathbf{p}_{\text{cart}} = [x, y]^\top$ is used as the EE target. Drake's `InverseKinematics` solver finds joint angles $\mathbf{q}_d = [q_{d,1}, q_{d,2}]^\top$ such that:

$$f_{\text{FK}}(\mathbf{q}_d) = \mathbf{p}_{\text{cart}}, \quad \|\mathbf{q}_d - \mathbf{q}_{\text{prev}}\|_2 \rightarrow \min$$

where f_{FK} is the forward-kinematics map of the manipulator and warm-starting with the previous solution ensures continuity. The desired joint velocities $\dot{\mathbf{q}}_d$ are computed numerically from consecutive IK solutions.

Block 6 — Joint-Space Computed-Torque Controller

A computed-torque law cancels the manipulator nonlinear dynamics and imposes a linear PD law on the joint error:

$$\boldsymbol{\tau} = M(\mathbf{q}) \ddot{\mathbf{q}}_d + C(\mathbf{q}, \dot{\mathbf{q}}) \dot{\mathbf{q}} + G(\mathbf{q}) + K_p(\mathbf{q}_d - \mathbf{q}) + K_d(\dot{\mathbf{q}}_d - \dot{\mathbf{q}})$$

- **Inputs:** desired state $[\mathbf{q}_d, \dot{\mathbf{q}}_d, \ddot{\mathbf{q}}_d]$ and current state $[\mathbf{q}, \dot{\mathbf{q}}]$
- **Output:** $\boldsymbol{\tau} \in \mathbb{R}^2$ (joint torques)
- **Gains:** $K_p = 200 \text{ N} \cdot \text{m}/\text{rad}$, $K_d = 60 \text{ N} \cdot \text{m} \cdot \text{s}/\text{rad}$, $\tau_{\text{max}} = 100 \text{ N} \cdot \text{m}$

4 Block Diagram

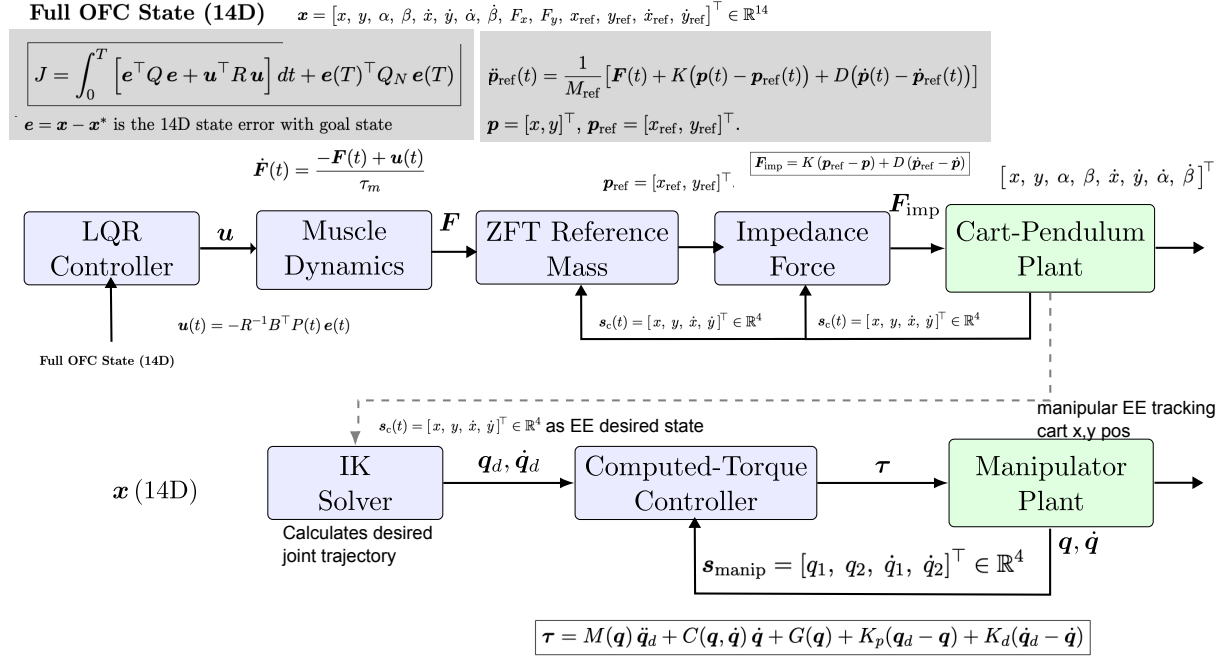


Figure 1: Block diagram of the `lqr-applied-to-cart-manip-following-cart` mode. Solid arrows: signals in the primary control path. Dashed arrows: auxiliary feedback/feedforward signals.

5 Signal Flow Summary

Table 1 summarises every signal connection in the diagram.

Table 1: Signal connections in the control loop.

From	To	Signal	Dim
LQR Controller	Muscle Dynamics	$\mathbf{u}$ (neural command)	$\mathbb{R}^2$
Muscle Dynamics	ZFT Reference Mass	$\mathbf{F}$ (muscle force)	$\mathbb{R}^2$
Muscle Dynamics	State Mux (14D)	$\mathbf{F}$ (for LQR state)	$\mathbb{R}^2$
Cart-Pendulum Plant	ZFT Reference Mass	$\mathbf{s}_{\text{cart}}$	$\mathbb{R}^4$
Cart-Pendulum Plant	Impedance Force	$\mathbf{s}_{\text{cart}}$	$\mathbb{R}^4$
ZFT Reference Mass	Impedance Force	$\mathbf{s}_{\text{ref}}$ (<i>only ZFT output goes to impedance, not $\mathbf{F}$</i>)	$\mathbb{R}^4$
Impedance Force	Cart-Pendulum Plant	$\mathbf{F}_{\text{imp}}$ (actuation)	$\mathbb{R}^2$
Cart-Pendulum Plant	State Mux	$\mathbf{s}_{\text{cart}}$ (8D)	$\mathbb{R}^8$
Muscle Dynamics	State Mux	$\mathbf{F}$	$\mathbb{R}^2$
ZFT Reference Mass	State Mux	$\mathbf{s}_{\text{ref}}$	$\mathbb{R}^4$
State Mux	ZOH (state hold)	$\mathbf{x}$ (14D)	$\mathbb{R}^{14}$
ZOH	LQR Controller	$\mathbf{x}$ (14D, held)	$\mathbb{R}^{14}$
Cart-Pendulum Plant	IK Solver	$[x, y, \dot{x}, \dot{y}]$	$\mathbb{R}^4$
IK Solver	Computed-Torque Ctrl.	$[\mathbf{q}_d, \dot{\mathbf{q}}_d, \ddot{\mathbf{q}}_d]$	$\mathbb{R}^6$
Manipulator Plant	Computed-Torque Ctrl.	$[\mathbf{q}, \dot{\mathbf{q}}]$	$\mathbb{R}^4$
Computed-Torque Ctrl.	Manipulator Plant	$\boldsymbol{\tau}$ (joint torques)	$\mathbb{R}^2$

6 Linearised System

The LQR is designed on the linearised 14D OFC system. Let $\mathbf{x}$ be the full state and $\mathbf{u}$ the neural command. Around the equilibrium ($\mathbf{x}^*, \mathbf{u}^* = \mathbf{0}$) the dynamics are:

$$\dot{\mathbf{x}} = A \mathbf{x} + B \mathbf{u}$$

The $A \in \mathbb{R}^{14 \times 14}$ and $B \in \mathbb{R}^{14 \times 2}$ matrices encode:

- **Cart-Pendulum linearisation:** obtained from the Drake `Linearize()` helper applied to the 8D `MultibodyPlant` at the upright equilibrium.
- **Muscle augmentation:** the muscle dynamics $\dot{F}_i = (-F_i + u_i)/\tau_m$ extend the A and B matrices by two rows/columns each.
- **ZFT augmentation:** the reference-mass dynamics extend the state by four more rows.

The discrete-time Riccati equation is solved backward in time from $t = T$ to $t = 0$ at timestep $\Delta t = 10$ ms to obtain $P(t)$, then the gain schedule $K(t) = R^{-1} B^\top P(t)$ is stored and interpolated online.

7 Simulation Parameters

Table 2: Default simulation parameters.

Parameter	Value	Description
Cart mass	3.0 kg	
Pendulum mass	0.3 kg	
Pendulum length	0.25 m	
Cart damping	0.5 N · s/m	
Pendulum damping	0.1 N · s/rad	
τ_m	0.03 s	Muscle time constant
M_{ref}	1.0 kg	ZFT reference mass
K	50 N/m	Impedance stiffness
D	10 N · s/m	Impedance damping
K_p	200 N · m/rad	Joint PD: position gain
K_d	60 N · m · s/rad	Joint PD: velocity gain
LQR horizon T	10.0 s	
Δt (LQR)	10 ms	Riccati integration step
Simulation Δt	1 ms	Drake integrator step

8 Implementation Notes

- A **Zero-Order Hold (ZOH)** at 10 ms is inserted between the 14D state multiplexer and the LQR input to break the algebraic loop introduced by directly feeding the LQR output (through muscle and impedance) back to itself via the ZFT state.
- The **IK solver** is warm-started with the previous joint solution at every step to ensure smooth, continuous joint trajectories and avoid large discontinuous jumps.
- All systems are **Drake LeafSystems** wired inside a single **DiagramBuilder**, which guarantees correct simulation semantics (context-based state isolation, publication order, etc.).
- The cart-pendulum and manipulator share the **same MultibodyPlant** but have separate **ModelInstance** handles, enabling independent state access and actuation while capturing any physical coupling.

9 Equation Flow Summary

This section presents the complete signal chain as a single, self-contained narrative — from sensor reading to actuator output — showing exactly how each equation feeds into the next at every simulation step.

Step-by-step signal chain

At every simulation step t , the following sequence executes:

(1) Read plant state. The cart-pendulum plant outputs its 8D state:

$$\mathbf{s}_{\text{cart}}(t) = [x, y, \alpha, \beta, \dot{x}, \dot{y}, \dot{\alpha}, \dot{\beta}]^\top \in \mathbb{R}^8$$

A demultiplexer splits this into four 2-vectors $[\mathbf{p}, \boldsymbol{\theta}, \dot{\mathbf{p}}, \dot{\boldsymbol{\theta}}]$, and a multiplexer re-assembles the *cart-only* sub-state:

$$\mathbf{s}_c(t) = [x, y, \dot{x}, \dot{y}]^\top \in \mathbb{R}^4$$

(2) Muscle dynamics — low-pass filter on neural command. The LQR command $\mathbf{u} \in \mathbb{R}^2$ (from the previous ZOH slot) drives first-order actuator dynamics:

$$\dot{\mathbf{F}}(t) = \frac{-\mathbf{F}(t) + \mathbf{u}(t)}{\tau_m} \implies \mathbf{F}(t) \in \mathbb{R}^2$$

The muscle state $\mathbf{F}$ is (a) fed to the **ZFT reference mass only** (not to the impedance block directly), and (b) stacked into the 14D state for the LQR. $\mathbf{F}$ reaches the impedance output only indirectly, by shaping how the reference mass moves.

(3) ZFT reference mass — smooth reference trajectory. The virtual reference mass is driven by muscle force and the error to the cart:

$$\ddot{\mathbf{p}}_{\text{ref}}(t) = \frac{1}{M_{\text{ref}}} [\mathbf{F}(t) + K(\mathbf{p}(t) - \mathbf{p}_{\text{ref}}(t)) + D(\dot{\mathbf{p}}(t) - \dot{\mathbf{p}}_{\text{ref}}(t))]$$

Integrating gives:

$$\mathbf{s}_{\text{ref}}(t) = [x_{\text{ref}}, y_{\text{ref}}, \dot{x}_{\text{ref}}, \dot{y}_{\text{ref}}]^\top \in \mathbb{R}^4$$

(4) Impedance force — virtual spring-damper actuation. Using only $\mathbf{s}_c$ (actual cart state) and $\mathbf{s}_{\text{ref}}$ (ZFT output), the impedance block computes the external force applied to the cart:

$$\mathbf{F}_{\text{imp}}(t) = K(\mathbf{p}_{\text{ref}} - \mathbf{p}) + D(\dot{\mathbf{p}}_{\text{ref}} - \dot{\mathbf{p}}) \in \mathbb{R}^2$$

Note: $\mathbf{F}$ (muscle force) is *not* a direct input here. It enters this equation only indirectly through $\mathbf{p}_{\text{ref}}$, which was shaped by the ZFT ODE in Step 3. This is the *sole actuation input* to the cart-pendulum plant. The pendulum angles α, β are shaped indirectly by the resulting cart acceleration.

(5) Full 14D state assembly. All sub-states are concatenated:

$$\mathbf{x}(t) = \underbrace{[x, y, \alpha, \beta]}_{\text{cart pos.}} \oplus \underbrace{[\dot{x}, \dot{y}, \dot{\alpha}, \dot{\beta}]}_{\text{cart vel.}} \oplus \underbrace{[F_x, F_y]}_{\text{muscle}} \oplus \underbrace{[x_{\text{ref}}, y_{\text{ref}}, \dot{x}_{\text{ref}}, \dot{y}_{\text{ref}}]}_{\text{ZFT ref.}} \in \mathbb{R}^{14}$$

A Zero-Order Hold (ZOH, $\Delta = 10$ ms) samples this to $\hat{\mathbf{x}}(t)$ before passing it to the LQR.

(6) Finite-horizon LQR — optimal neural command. The state error $e(t) = \hat{x}(t) - x^*$ enters the time-varying gain computed by the backward Riccati pass:

$$u(t) = - \underbrace{R^{-1}B^\top P(t)}_{K(t)} e(t) \in \mathbb{R}^2$$

This $u(t)$ closes the outer loop back to **Step 2** (muscle input) at the next time step.

(7) IK solver — cart position to joint angles. In parallel, the cart position $[x, y]$ is used as the EE target for the manipulator’s inverse kinematics:

$$q_d(t) = f_{\text{IK}}([x(t), y(t)], q_d(t-\Delta t)) \in \mathbb{R}^2$$

Desired joint velocity is approximated numerically: $\dot{q}_d = (q_d(t) - q_d(t-\Delta t))/\Delta t$.

(8) Computed-torque controller — joint torques. The nonlinear manipulator dynamics are cancelled and a PD law is imposed:

$$\tau(t) = M(q)\ddot{q}_d + C(q, \dot{q})\dot{q} + G(q) + K_p(q_d - q) + K_d(\dot{q}_d - \dot{q}) \in \mathbb{R}^2$$

τ is applied directly to the manipulator joints, driving the end-effector to follow the cart trajectory.

Compact causal chain

The complete equation flow can be read as a single chain of dependencies:

$$\begin{array}{l} \underbrace{s_{\text{cart}}}_{\text{plant}} \xrightarrow{\text{demux/mux}} s_c \xrightarrow[\text{ZFT}]{F} s_{\text{ref}} \xrightarrow{\text{imp.}} F_{\text{imp}} \longrightarrow \text{cart plant} \\ s_{\text{cart}} \oplus F \oplus s_{\text{ref}} \xrightarrow{\text{mux}} x_{14} \xrightarrow{\text{ZOH}} \hat{x} \xrightarrow{\text{LQR}} u \xrightarrow{\tau_m} F \quad (\text{closes loop}) \\ s_c \xrightarrow{\text{IK}} q_d, \dot{q}_d \xrightarrow{\text{CTC}} \tau \longrightarrow \text{manip. plant} \end{array}$$

Each arrow represents a Drake system connection. The cart-pendulum loop (top two lines) and the manipulator loop (bottom line) run concurrently inside the same Drake **Diagram** at every integration step.

10 Simulation Results

Figure 2 shows the full result dashboard produced by a representative run with target $(x^*, y^*) = (-1.0, 0.5)$ m, duration 10 s, and the default parameters from Table 2. The dashboard is divided into five rows, each capturing a different aspect of the closed-loop behaviour.

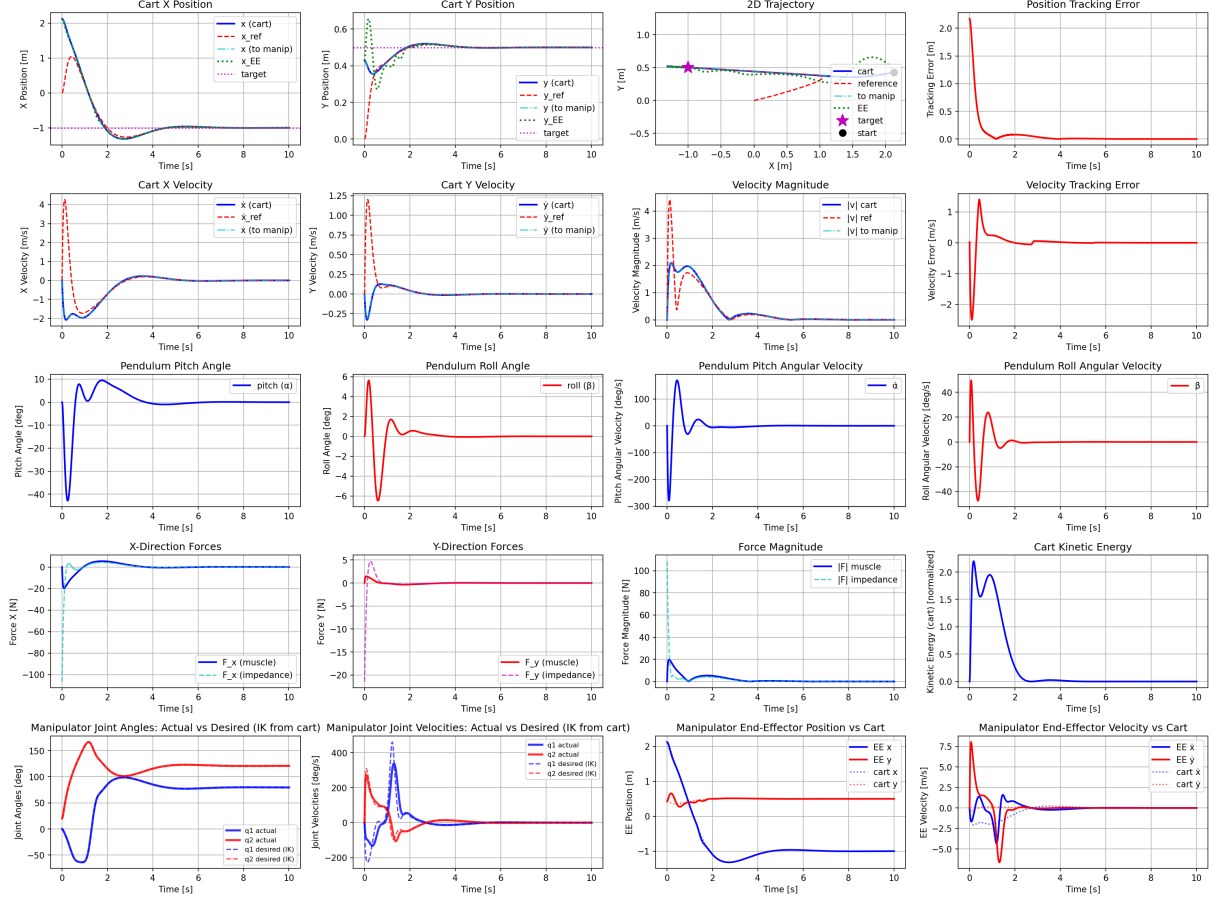


Figure 2: Full result dashboard for the `lqr-applied-to-cart-manip-following-cart` mode. Target: $(x^*, y^*) = (-1.0, 0.5)$ m, horizon $T = 10$ s, duration 10 s.

Row 1 — Cart Position and 2D Trajectory

Col 1 — Cart X Position Compares the actual cart $x(t)$ (blue solid), the ZFT reference $x_{\text{ref}}(t)$ (red dashed), the trajectory sent to the manipulator (cyan dash-dot), and the end-effector position $x_{\text{EE}}(t)$ (green dotted). The ZFT reference leads the cart slightly because the reference mass responds to the muscle force before the cart does. The magenta horizontal line is the target x^* .

Col 2 — Cart Y Position Same legend as Col 1 but for the y -axis. Convergence to $y^* = 0.5$ m is visible within the first few seconds.

Col 3 — 2D Trajectory XY-plane view of all four trajectories. The black circle marks the initial position; the magenta star is the target. This subplot shows how closely the EE (green dotted) tracks the cart (blue solid) while both converge toward the goal.

Col 4 — Position Tracking Error Euclidean distance $\|\mathbf{p}(t) - \mathbf{p}_{\text{ref}}(t)\|_2$ between the cart and the ZFT reference. A smaller error means the impedance spring-damper is keeping the

cart tightly coupled to the reference mass.

Row 2 — Cart Velocity

Col 1 — Cart X Velocity Actual cart velocity $\dot{x}$ vs. ZFT reference velocity $\dot{x}_{\text{ref}}$ and the velocity sent to the manipulator IK. Peak velocity occurs during the initial transient as the LQR drives the system toward the goal.

Col 2 — Cart Y Velocity Same as Col 1 for the y -direction.

Col 3 — Velocity Magnitude $|\mathbf{v}|$ for cart, reference, and manipulator command. The smooth peak-and-decay profile reflects the finite-horizon LQR policy, which front-loads effort and then coasts to rest.

Col 4 — Velocity Tracking Error $|\dot{\mathbf{p}}| - |\dot{\mathbf{p}}_{\text{ref}}|$. Small magnitude confirms that the impedance coupling keeps the cart velocity well synchronised with the reference.

Row 3 — Pendulum Angles and Angular Velocities

Col 1 — Pendulum Pitch α Pitch angle of the pendulum over time. Deviations from zero indicate the pendulum is disturbed by cart acceleration; the LQR's high pendulum weights ($Q_{33} = Q_{44} = 500$) work to suppress these oscillations.

Col 2 — Pendulum Roll β Roll angle. Because the target has a non-zero y -component, the y -direction acceleration excites roll momentarily before settling.

Col 3 — Pitch Angular Velocity $\dot{\alpha}$ Rate of change of pitch. Rapid changes correspond to moments of high cart acceleration; the system damps these through the CTC and impedance gains.

Col 4 — Roll Angular Velocity $\dot{\beta}$ Rate of change of roll. Symmetric behaviour to $\dot{\alpha}$ but driven by y -direction dynamics.

Row 4 — Forces

Col 1 — X-Direction Forces Muscle force F_x (solid blue) and impedance force $F_{x,\text{imp}}$ (dashed cyan). The muscle force is the neural command after low-pass filtering ($\tau_m = 0.03$ s). The impedance force is the spring-damper output that actually actuates the cart; it closely follows the muscle force with a small lag caused by the ZFT ODE.

Col 2 — Y-Direction Forces Same as Col 1 for F_y (red solid) and $F_{y,\text{imp}}$ (magenta dashed).

Col 3 — Force Magnitude $\|\mathbf{F}\|$ for both muscle and impedance. The peak at $t \approx 0$ is the initial burst from the LQR; the gradually decreasing envelope reflects the finite-horizon policy reducing effort as the horizon shortens.

Col 4 — Cart Kinetic Energy Proxy metric $\frac{1}{2}(\dot{x}^2 + \dot{y}^2)$ (unnormalised by mass). This peaks during the transient and decays to near-zero at the target, confirming the cart comes to rest.

Row 5 — Manipulator State

Col 1 — Joint Angles: Actual vs. Desired q_1 and q_2 (solid) plotted against IK-desired angles $q_{d,1}$ and $q_{d,2}$ (dashed). Close tracking here indicates the CTC is successfully following the IK output; residual error is kept small by the high PD gains ($K_p = 200$ N · m/rad).

- Col 2 — Joint Velocities: Actual vs. Desired** $\dot{q}_1, \dot{q}_2$ vs. IK-desired $\dot{q}_{d,1}, \dot{q}_{d,2}$. Velocity tracking is inherently noisier than position tracking because $\dot{q}_d$ is computed by finite differencing consecutive IK solutions.
- Col 3 — EE Position vs. Cart** Manipulator end-effector coordinates x_{EE}, y_{EE} (solid) vs. actual cart coordinates x, y (dotted). Good overlap throughout the simulation confirms the EE successfully follows the cart, which is the objective of the manipulator control loop.
- Col 4 — EE Velocity vs. Cart Velocity** $\dot{x}_{EE}, \dot{y}_{EE}$ vs. $\dot{x}, \dot{y}$. Velocity tracking is slightly lagged because the IK and CTC introduce a one-step delay; this is visible as a small phase offset during the transient peak.